

Water Policy, Laws, and Global Resource Governance

UN Sustainable Development Goal 6 (SDG 6) and Human Rights

United Nations Sustainable Development Goal 6 (SDG 6) commits the global community to ensuring access to safe water and sanitation for all by 2030. SDG 6 targets encompass universal drinking water access, adequate sanitation, improved ambient water quality, increased water-use efficiency, integrated water management, and water ecosystem protection.

In 2010, the UN General Assembly explicitly recognized access to clean, safe, acceptable, and affordable drinking water and sanitation as a fundamental human right essential for the full enjoyment of life and human dignity.

Transboundary Water Treaties and International Water Law

Over 260 international river basins cross political boundaries, serving 40 percent of the world's population. Transboundary water governance requires international legal treaties and river basin commissions to prevent hydropolitical conflict.

International water law principles—codified in the 1997 UN Watercourses Convention—rely on two foundational pillars: Equitable and Reasonable Utilization of shared waters, and the Obligation Not to Cause Significant Harm to downstream riparian states.

Economic Instruments: Water Pricing, Water Rights, and Subsidies

Economic instruments incentivize water conservation while funding utility infrastructure maintenance. Block-rate tariffs charge higher volumetric rates for high-volume consumption, protecting basic lifeline water affordability for low-income households while discouraging wasteful use.

Tradable water rights allocation markets allow agricultural and industrial water users to trade volumetric water entitlements, encouraging water-efficient technology adoption and transferring water to high-value uses during extreme scarcity.

Community-Based Watershed Management and Water Governance

Effective water governance requires decentralization and direct community participation. Water User Associations (WUAs) empower local farming communities to manage canal distribution, maintain infrastructure, and resolve local allocation disputes equitably.

Integrating traditional ecological knowledge—such as ancient qanat underground channels, stepwells, and traditional rainwater harvesting bunds—with modern hydrologic science yields culturally robust, climate-resilient water stewardship.